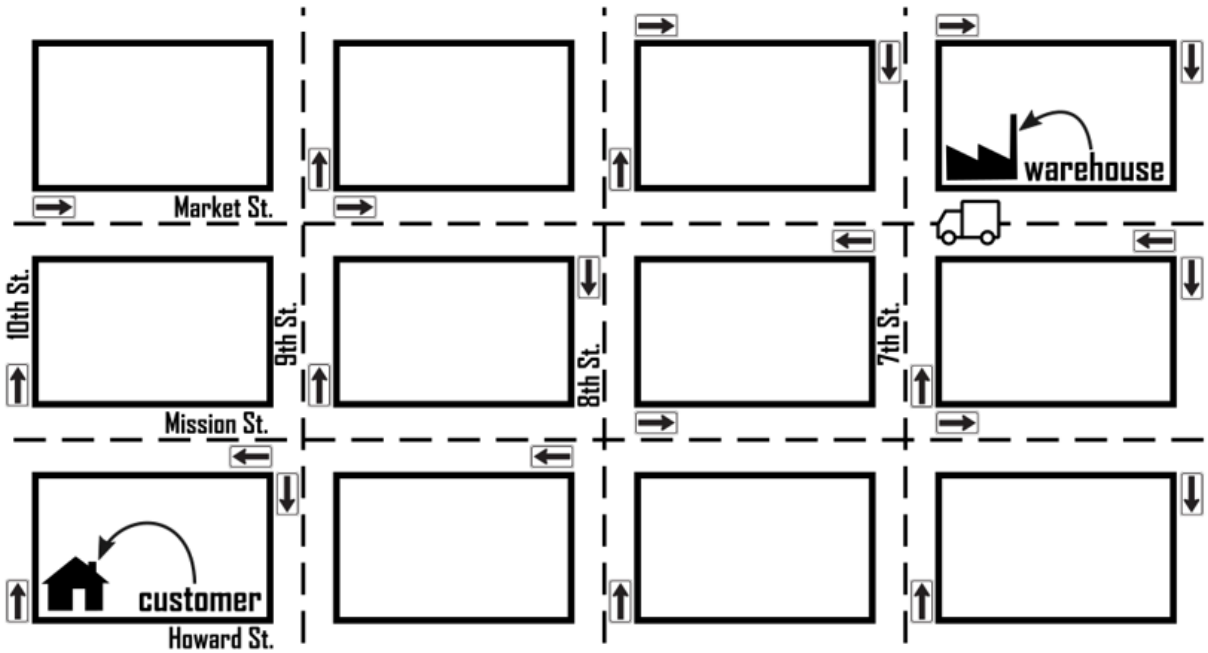


Figure 14.8 An example map: a portion of San Francisco’s downtown, where blocks form a regular grid.

We’ll develop a generic solution for this problem incrementally, across this and the next few sections, starting with a furtherly simplified scenario and removing those simplifications step by step, while presenting more complex algorithms to solve these cases.

So, to start our discussion, we need to think about our goal: what’s the “best” route? We can assume, for instance, that the best route is the shortest route, but we might as well prefer to find the fastest route, or the cheapest one, depending on our requirements.

For the moment, let’s assume we want the shortest route: if we simplify the scenario and ignore factors like traffic, conditions of the roads, speed limits, then we can